



1ACC0221 – BIG DATA

Sección: 18516

Program: Computer Science

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1. Project Proposal:

This project is developed within the ecosystem of open-source scientific artificial intelligence tools. This domain was selected due to the accelerated and continuous growth of public repositories related to models, libraries, frameworks, and research-oriented tools. According to GitHub's Octoverse 2024 report, during 2024 developers created more than 70,000 new public and open-source generative AI projects, while total contributions to this type of project increased by nearly 60%. This growth reflects a highly dynamic, broad, and increasingly difficult environment to explore in a structured way.

In this context, the main problem lies in the fact that the AI software ecosystem is growing in an explosive and disorganized manner. Every week, new GitHub repositories appear related to scientific papers, experimental implementations, or applied tools, but there is no structured system that allows users to discover, compare, and understand which repositories are truly influential beyond superficial metrics such as stars or forks. This creates difficulties in identifying similar repositories, recognizing thematic families, and prioritizing relevant tools within an increasingly large universe.

Based on this, the proposed system seeks to answer four specific questions:

- Which AI repositories exist within a specific domain, and which of them are similar to each other?
- To which thematic family or cluster does a given repository belong?
- What other tools should someone explore if they are already using a specific tool?
- Which repositories are structurally the most influential within the ecosystem, beyond their apparent popularity?

The dataset is suitable for the course because it combines free text in descriptions and topics, numerical variables such as stars and forks, temporal variables such as age and recent activity, and implicit relationships between repositories. This combination supports the

subsequent stages of the project, including representation, clustering, recommendation, and graph analysis.

2. Data Sources and Access:

2.1. Source Inventory:

Source 1: GitHub API

URL: <https://api.github.com>

License: Public API under GitHub Terms of Service.

Format: JSON via REST API, processed with PyGithub 2.x.

Current size: 86 repositories and 13 columns, with projected growth to 500–1000 repositories in later iterations.

Source 2: Papers With Code

URL: <https://paperswithcode.com/api/v1>

License: Public API with no authentication required.

Format: JSON via REST API.

Intended use: To enrich each repository with its associated scientific paper, the task it addresses, and the reported benchmarks.

Source 3: Hugging Face Hub

URL: <https://huggingface.co/>

License: Public API with official Python SDK support.

Format: JSON.

Intended use: A tertiary source to enrich the catalog with models and datasets linked to the analyzed repositories.

2.2. Ethics and Access Note:

The data used in this project comes from public sources accessible through official APIs. At this stage, the main source is the GitHub API, which is accessed using a personal token only for authentication and rate-limit control.

No scraping of private profiles was performed, and no information restricted by third-party authentication was accessed. The extraction is limited to technical metadata from public repositories,

such as descriptions, topics, primary language, stars, forks, and activity dates.

No real names, email addresses, private messages, or sensitive content from individual users were collected. Therefore, the risk related to personal data is minimal. In addition, the raw dataset will not be redistributed outside the team, and the processed artifacts will be used exclusively for academic purposes within the course.

3. Dataset Design and Construction:

3.1. Schema Draft:

Column	Type	Description	Key
repo_id	integer	Unique identifier of the repository on GitHub	PK
nombre	string	Full name in <code>user/repository</code> format	FK → papers, dependencies
descripcion	string	Short description provided by the author	
lenguaje	string	Main programming language	
stars	integer	Number of accumulated stars	
forks	integer	Number of repository forks	
topics	list	Thematic tags declared by the author	

fecha_creacion	datetime	Repository creation date	
ultimo_commit	datetime	Date of the last recorded commit	

3.2. Relaciones y joins esperados:

Table A	Table B	Join Field	Description
repos	papers	nombre	Links each repository with its related scientific paper.
repos	dependencies	repo_id	Links each repository with its declared libraries or dependencies.
repos	topics_expanded	repo_id	Normalizes topics into an expanded table for further analysis.

4. Diccionario de datos:

Column	Type	Description	Range or Observed Values
repo_id	integer	Unique GitHub ID	7 digits or more
nombre	string	<code>user/repo</code>	Example: <code>microsoft/qlib</code>
descripcion	string	Free text provided by the author	1 to ~200 characters, “No description” if empty
lenguaje	string	Main language	Python, Jupyter Notebook, C++, TeX, Unknown
stars	integer	Superficial popularity	51 — 41,080 in the current dataset
forks	integer	Number of times it was forked	4 — 8,394
topics	list	Author-defined tags	Between 2 and 20 per repository
fecha_creacion	datetime	Creation date	Repositories with up to 3,714 days of age
ultimo_commit	datetime	Last recorded push	From today up to 3,055 days ago
url	string	Direct link	Always <code>https://github.com/...</code>
dias_sin_commit	integer	Recent inactivity	0 (active today) — 3,055 days
antiguedad_dias	integer	Repository age	402 — 3,714 days
num_topics	integer	Tag richness	2 — 20

5. Exploratory Analysis and Dataset Scale:

The key figures of the current dataset are as follows:

Total repositories: 86

Total columns: 13

Missingness after cleaning: 0 across all columns

Memory estimate: 38.4 KB

Stars range: 51 to 41,080, with a median of 1,241

This distribution is strongly right-skewed, which is typical in GitHub repositories where a small number of projects accumulate most of the stars.

Unique languages: 7

Python is the dominant language with 38 repositories, followed by 29 repositories with no declared language (documentation or notebooks not classified by GitHub).

Repositories active in the last 30 days: 16 out of 86

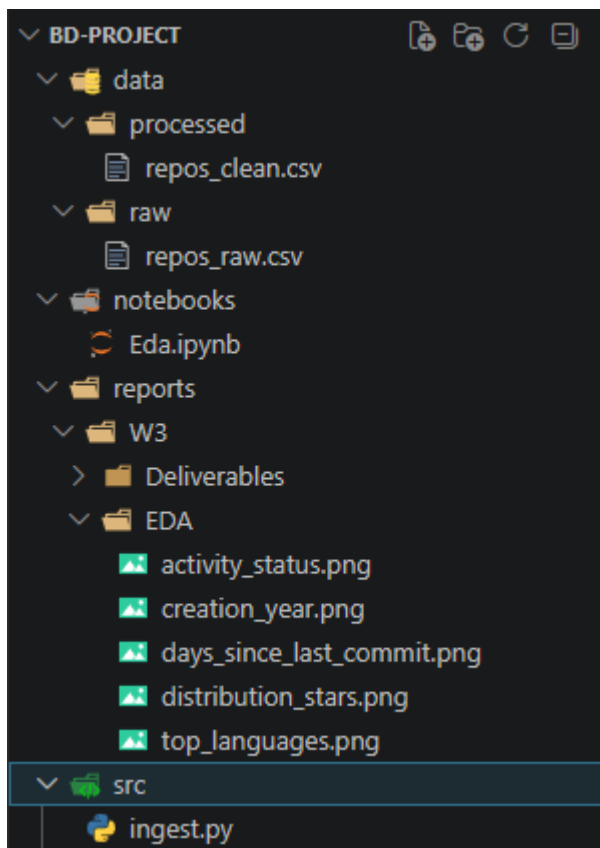
This suggests that most of the repositories correspond to research projects that are published but do not receive continuous maintenance.

Average age: 2,245 days (approximately 6 years)

This indicates that the dataset captures established tools within the ecosystem.

The chart [reports/scale_analysis.png](#) shows the log-scale distribution of stars, the ranking of programming languages, and the distribution of days since the last commit.

6. One processed dataset V1:



The first processed version of the dataset is generated through a pipeline organized in stages.

Step 1: the GitHub API is queried to collect the metadata of the repositories defined in the project queries.

Step 2: the result of this extraction is stored as a raw dataset in the path `data/raw/repos_raw.csv`.

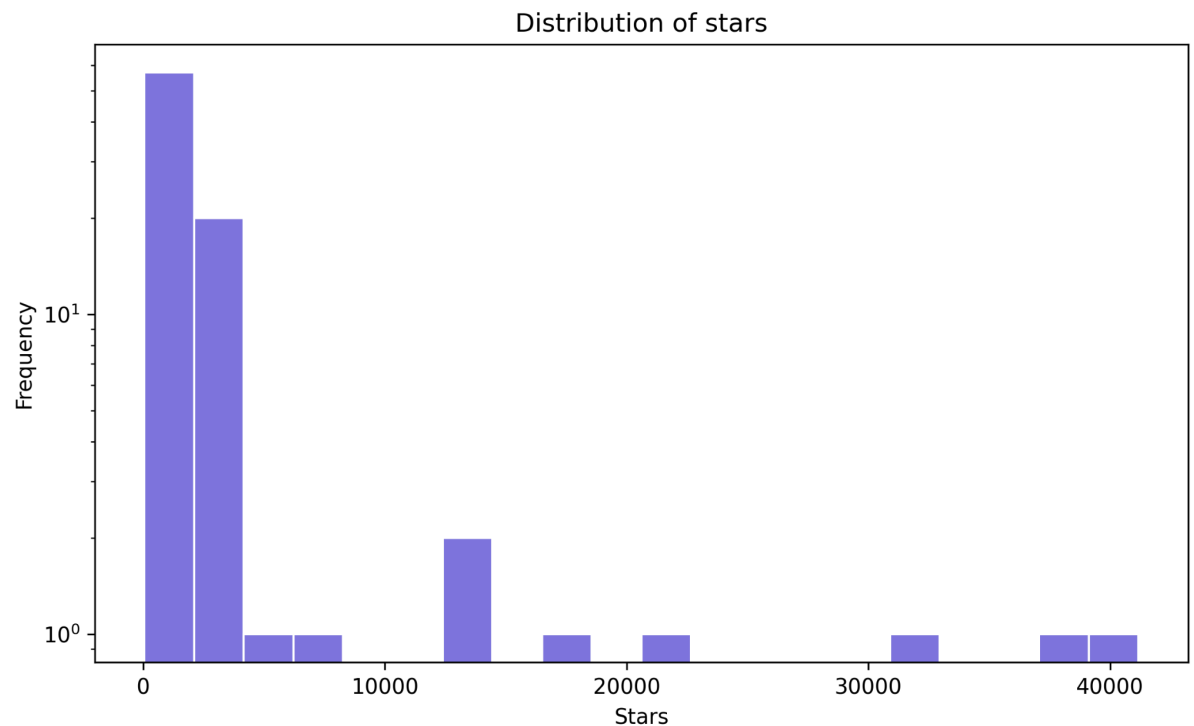
Step 3: this raw file is then subjected to a cleaning process, which includes handling missing values, normalizing variables, and creating new features such as `dias_sin_commit`, `antiguedad_dias` and `num_topics`.

Step 4: Finally, the result is saved as a processed dataset in the path `data/processed/repos_clean.csv`

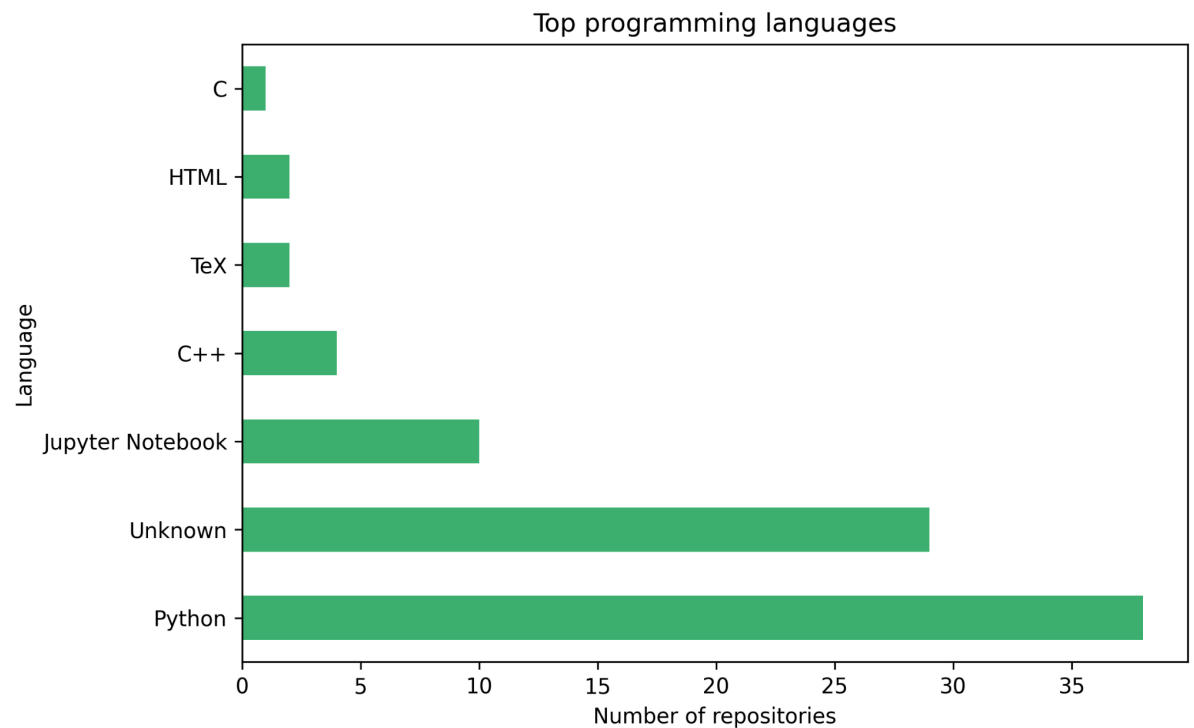
El pipeline puede ejecutarse desde la raíz del proyecto con el siguiente comando:

```
python src/ingest.py
```

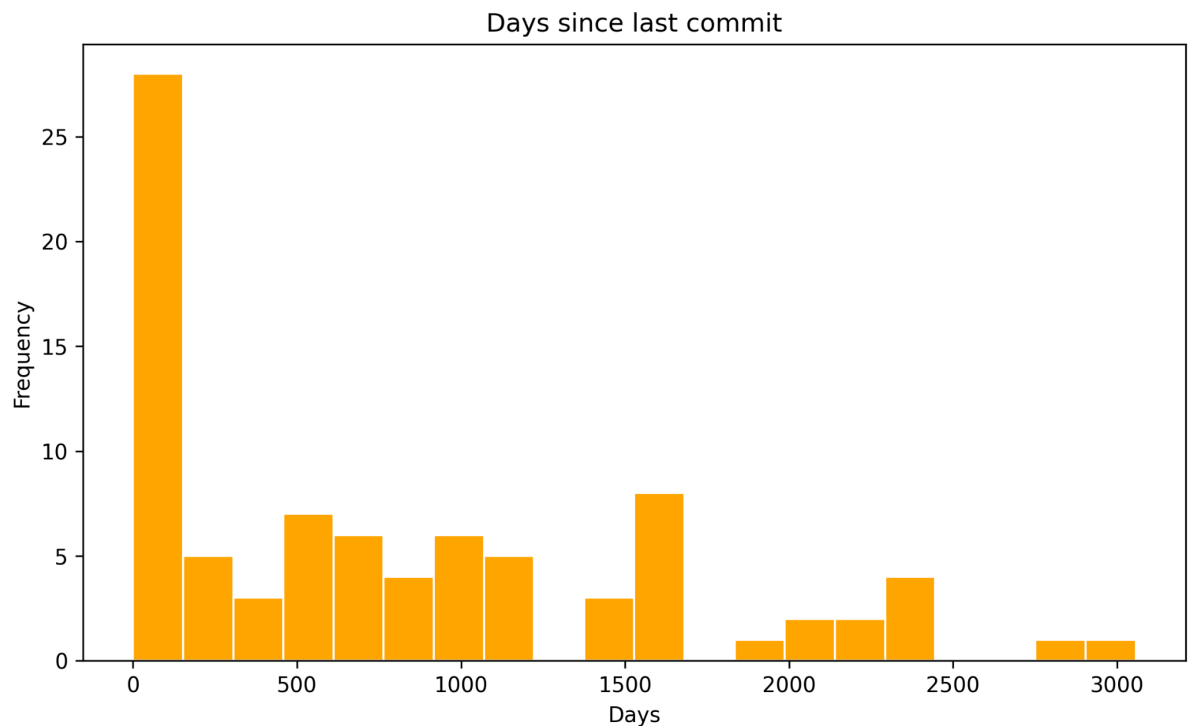
7. EDA Analysis:



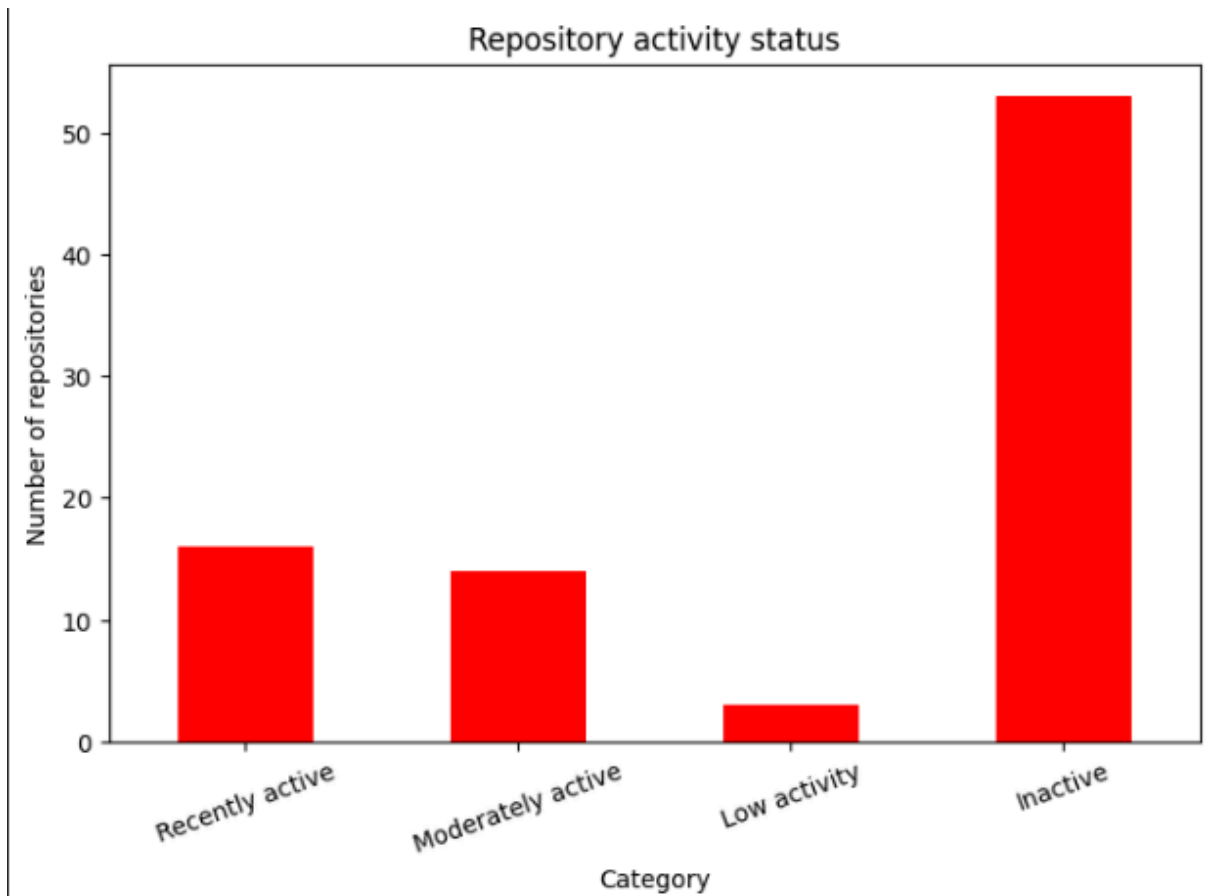
The distribution of stars shows a strong right-skewed pattern. Most repositories have low or intermediate levels of popularity, while only a small group concentrates very high numbers of stars. This indicates that visibility within the dataset is not distributed uniformly.



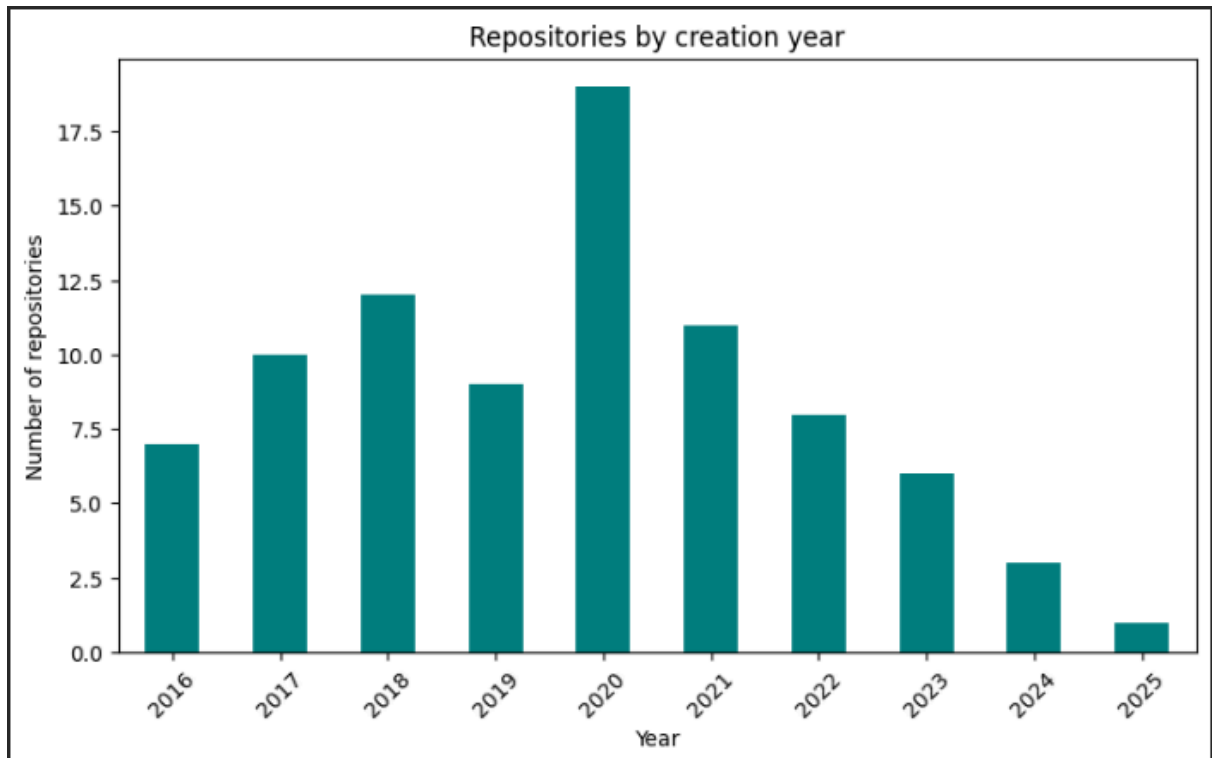
The programming languages chart shows a clear predominance of Python within the dataset, followed by a significant number of repositories with no declared language and by Jupyter Notebook. This is consistent with the type of ecosystem analyzed, since Python is the most widely used language in artificial intelligence and scientific software projects.



The distribution of days since the last commit shows that some repositories still exhibit recent activity, but there is also a considerable number of projects with long periods without updates. This suggests that the dataset includes both active tools and repositories with limited or discontinued maintenance.



The activity status classification shows that the predominant category is inactive repositories. To a lesser extent, there are repositories with recent, moderate, and low activity. This result indicates that the collected ecosystem combines recognized projects with different levels of recent maintenance.



The analysis by year of creation shows that the dataset has a higher concentration of repositories created between 2018 and 2021, with 2020 standing out as the year with the highest frequency in the analyzed sample.

8. Bibliography:

GitHub Staff. *Octoverse 2024: AI leads Python to top language as the number of global developers surges*. GitHub Blog, 2024.